Customer Segmentation & Prediction

Business Recommendations Report

Week 11 Advanced Machine Learning Capstone

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1. Executive Summary

This report presents actionable business recommendations derived from a machine-learning analysis of 500 customers in a telecommunications dataset. The overall churn rate is **10.6%**. Using K-Means clustering (K=6, silhouette score=0.3080), customers were segmented into **6 distinct groups**, each with unique behavioural characteristics and churn risk levels. Separate Random Forest classifiers were trained for each segment, with hyperparameter tuning via GridSearchCV.

2. Dataset Overview

Property	Value
Total Customers	500
Features	Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitize
Target Variable	Churn (0=No, 1=Yes)
Overall Churn Rate	10.6%
Missing Values	0
Duplicate Rows	0
Number of Segments	6

3. Clustering Analysis Results

Three clustering algorithms were evaluated: K-Means (silhouette=0.3080), Agglomerative/Hierarchical (silhouette=0.2292), and DBSCAN (silhouette=N/A — 0 noise points, 1 clusters). K-Means was selected as the primary method because it produced the most interpretable and business-relevant cluster boundaries.

Cluster	Segment Name	Count	%	Avg Tenure	Avg $/mo	Churn %	Contract
0	Loyal Long-Term Customers	91	18.2%	54.7 mo	$91.3	0.0%	Month-to-month
1	Premium High-Spenders	93	18.6%	53.1 mo	$159.5	0.0%	One year

2	High-Risk Churners	88	17.6%	15.0 mo	$126.8	26.1%	Month-to-month
3	Budget Conscious Customers	81	16.2%	49.6 mo	$60.8	0.0%	One year
4	Mid-Value Segment 1	69	13.8%	17.3 mo	$63.6	15.9%	Month-to-month
5	At-Risk Mid-Tenure Customers	78	15.6%	23.3 mo	$169.5	24.4%	One year

4. Segment-Specific Business Recommendations

Segment 2: High-Risk Churners

Customer Count: 88 customers (17.6% of total)

Avg Tenure: 15.0 months | **Avg Monthly Charges:** $126.8 | **Churn Rate:** 26.1%

Churn Observation: This segment has the highest churn rate among all groups. Customers here are most likely on month-to-month contracts and paying via electronic check — both correlated with higher churn in telecommunications.

Business Interpretation: These customers may be dissatisfied with service, price-sensitive, or exploring competitor offers. Losing them is costly in revenue and customer lifetime value.

Recommended Strategy: Immediate retention intervention: proactive outreach via phone or email within the next billing cycle. Offer contract upgrade incentives (e.g., discounted annual plan). Identify dissatisfied customers through CSAT surveys.

Marketing Approach: Targeted win-back campaigns. Loyalty discounts for contract extension. Highlight service reliability improvements.

Monitoring KPI: Monthly churn rate for this segment; contract upgrade conversion rate.

Model Performance (Tuned RF): Accuracy=0.8636 | F1=0.8 | AUC=1.0

Segment 5: At-Risk Mid-Tenure Customers

Customer Count: 78 customers (15.6% of total)

Avg Tenure: 23.3 months | **Avg Monthly Charges:** $169.5 | **Churn Rate:** 24.4%

Model Performance (Tuned RF): Accuracy=1.0 | F1=1.0 | AUC=1.0

Segment 0: Loyal Long-Term Customers

Customer Count: 91 customers (18.2% of total)

Avg Tenure: 54.7 months | **Avg Monthly Charges:** $91.3 | **Churn Rate:** 0.0%

Churn Observation: Highest average tenure with low churn rate. These customers have remained with the service for a long time.

Business Interpretation: The backbone of stable revenue. Their lifetime value is high. Even small improvements in their satisfaction can yield significant retention impact.

Recommended Strategy: Maintain satisfaction with excellent service quality. Reward loyalty with exclusive perks (early access, premium support, reward points).

Marketing Approach: Loyalty programme enrolment. Referral incentives (these customers are trusted brand advocates). Anniversary recognition messages.

Monitoring KPI: Net Promoter Score (NPS) for segment; referral rate.

Note on Prediction Model: Only one class present (all Churn=0). A classifier cannot be trained on a single-class problem. A separate classifier was not trained for this segment.

Segment 1: Premium High-Spenders

Customer Count: 93 customers (18.6% of total)

Avg Tenure: 53.1 months | **Avg Monthly Charges:** $159.5 | **Churn Rate:** 0.0%

Churn Observation: Highest average monthly charges. Revenue contribution per customer is significant.

Business Interpretation: Even a modest churn rate in this group has outsized revenue impact. These customers expect premium service quality.

Recommended Strategy: Dedicated account management. Proactive service reviews. Priority customer support line. Personalised offers.

Marketing Approach: Premium tier upsell and cross-sell (bundled services). Exclusive early-access to new features.

Monitoring KPI: Revenue retention rate; ARPU (Average Revenue Per User).

Note on Prediction Model: Only one class present (all Churn=0). A classifier cannot be trained on a single-class problem. A separate classifier was not trained for this segment.

Segment 3: Budget Conscious Customers

Customer Count: 81 customers (16.2% of total)

Avg Tenure: 49.6 months | **Avg Monthly Charges:** $60.8 | **Churn Rate:** 0.0%

Note on Prediction Model: Only one class present (all Churn=0). A classifier cannot be trained on a single-class problem. A separate classifier was not trained for this segment.

Segment 4: Mid-Value Segment 1

Customer Count: 69 customers (13.8% of total)

Avg Tenure: 17.3 months | **Avg Monthly Charges:** $63.6 | **Churn Rate:** 15.9%

Model Performance (Tuned RF): Accuracy=0.8889 | F1=0.6667 | AUC=0.8889

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5. Business Impact Estimation

Assumptions (clearly stated): The following impact estimates use only values derivable from the dataset. No revenue figures were fabricated.

The 'High-Risk Churners' segment has the highest churn rate (26.1%). With 88 customers at an average of $126.8/month, approximately 22 customers are at risk of churning. **Monthly revenue at risk: $2,790** (calculated as: at-risk customers × avg monthly charges). If a retention campaign retains even 30% of at-risk customers, this represents approximately $837/month in saved revenue. These are model estimates based on average charges and observed churn rates.

6. Limitations

1. Dataset size (500 customers) limits the robustness of segment-level models. Segments with few churners may show unreliable precision/recall. 2. The dataset does not include satisfaction scores, service tickets, or competitor pricing — factors known to influence churn. 3. Clustering is unsupervised and the number of segments is chosen by quantitative heuristics (silhouette score), not guaranteed business validity. 4. Revenue impact estimates assume the observed churn rate will persist; actual retention depends on campaign effectiveness. 5. Feature importance indicates model association, not causal relationships.

7. Conclusion

This analysis successfully segmented 500 customers into 6 distinct groups using K-Means clustering. Separate Random Forest classifiers were trained and tuned for each segment. The segment with the highest churn rate warrants immediate retention focus, while loyal long-term customers should be maintained with rewards and premium service. Budget and mid-value segments present upsell opportunities. All recommendations are grounded in actual data characteristics and model results.